

Aprendizagem Aplicada à Segurança

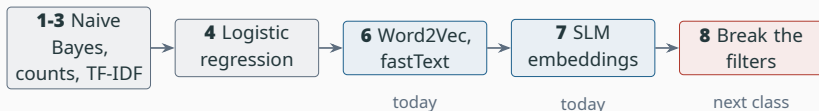
Class 3: SPAM Detection II, from Words to Meaning

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Where we stand



- Class 2 ended with a table of models within 0.94 to 0.96 F1, all seeing a message as **a bag of exact words**.
- Today: two questions. **Where do word and sentence vectors come from** (our data or pre-trained)? **How do we run a language model** we do not want inside the notebook?

Two blind spots of a bag of words

- **No similarity:** cash, money and prize are as unrelated as cash and banana.
- **No out-of-vocabulary handling:** fr33, fre3, freee are unknown words; their evidence is lost.

Message	Shared with claim your cash prize
collect your free money reward c4sh pr1ze, cl4im n0w	your only nothing

- A **vector** per word fixes the first: similar words get nearby vectors. **Sub-words** fix the second.

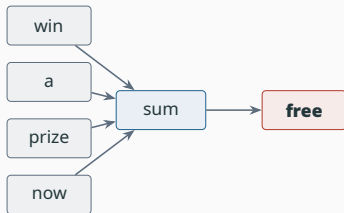
You shall know a word by the company it keeps

Words that occur in similar contexts have similar meanings. (distributional hypothesis, Harris 1954, Firth 1957)

- In spam, free appears next to entry, mobile, camera, txt, claim; in chat, home appears next to tonight, dinner, late.
- **Word2Vec** (Mikolov, Chen, et al. 2013) learns a dense vector per word (100 to 300 numbers) so that words with the **same neighbours** get the **same direction**.
- No labels needed: the text is the supervision. That matters, because **labels are scarce, text is cheap**.

Two ways to predict

CBOW: context $\rightarrow$ word



skip-gram: word $\rightarrow$ context



- **CBOW** averages the context to predict the word: fast, smooth for frequent words. **Skip-gram** predicts each context word from the word: better for rare words.

Training with negative sampling

For each observed pair (word w , context word c), push the score of the pair up and the score of k random words n_i down (Mikolov, Sutskever, et al. 2013):

$$\log \sigma(\mathbf{v}_w \cdot \mathbf{u}_c) + \sum_{i=1}^k \log \sigma(-\mathbf{v}_w \cdot \mathbf{u}_{n_i})$$

- $k = 5$ noise words instead of a softmax over the whole vocabulary: cost $O(k)$ per pair, not $O(|V|)$.
- Knobs: `vector_size`, `window(5)`, `min_count(2)`, `epochs(30)`, `sg`. Same logistic loss and descent as `lab 02_01`.

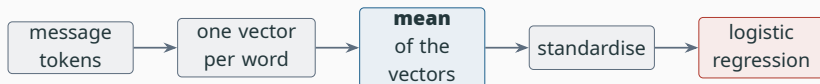
Vectors have geometry

Nearest neighbours by cosine similarity, vectors trained on **our** 65,000 tokens (skip-gram) and **Google News** (100 billion words):

word	in-domain (SMS)	pre-trained (news)
free	unlimited, camcorder, jamster, bluetooth	Free, FREE, complimentary
prize	caller, guaranteed, pound, bonus	prizes, award, grand_prize
home	lonely, chillin, knackered	house, homes, residence

- The news vectors know **English**. The in-domain vectors know **this corpus**: offers, phone-shop words, texting slang.

From words to a message



$$\mathbf{x} = \frac{1}{n} \sum_{i=1}^n \mathbf{v}_{w_i} \in \mathbb{R}^d$$

- Words without a vector are skipped; **word order is lost** (not free and free not are the same message).
- The classifier learns $d + 1$ weights instead of one per word: a much smaller problem than TF-IDF's thousands.

Two sources of vectors

	In-domain	Pre-trained
Corpus	the 4,095 training messages (65,000 tokens)	Google News: 100 billion words; Common Crawl (fastText)
Vocabulary Domain	3,107 words (min count 2) exactly SMS: slang, £, phone numbers	3 million words (we load 200,000) news and web pages
Cost	seconds on a CPU	1.7 GB download; 22 MB for the compressed fastText
Labels	none	none

- Only the **training** text is used, never the test messages, never the labels.
- Which is better for a spam filter? The intuition says *pre-trained*: 1.5 million times more text.

What a news vocabulary does not know about SMS

Spam tokens without a Google News vector, by frequency: num (512), ! (345), msg (40), pobox (30), optout (17), sae, mobileupd, freemsg.

Vectors	test tokens without a vector
Word2Vec in-domain	9.1%
fastText in-domain (n-grams)	0.0%
Word2Vec Google News (200k words)	7.1%
fastText Common Crawl, compressed	5.2%

- 13.5% of all **spam** tokens have no Google News vector: phone numbers, exclamation marks and SMS jargon **are** the spam signal.

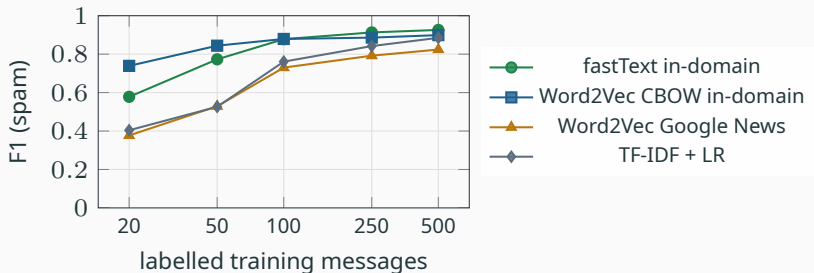
The result: in-domain wins

Mean of the word vectors + standardised logistic regression,
F1 on the 1,024 test messages:

Vectors	F1
fastText, in-domain	0.940
Word2Vec skip-gram, in-domain	0.935
Word2Vec CBOW, in-domain	0.912
Word2Vec Google News, pre-trained	0.888
fastText Common Crawl, pre-trained	0.882
TF-IDF + logistic regression (reference)	0.944

- Vectors learnt on the SMS beat vectors from 1.5 million times more text, and match TF-IDF with far fewer weights.

Few labels: unlabelled text is the resource

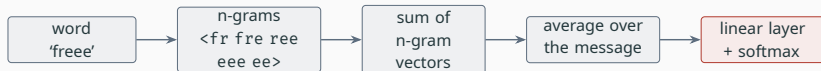


- The embeddings saw all the **unlabelled** training text; the classifier saw only n labels. TF-IDF has no unlabelled stage.

When does pre-training pay?

- **Coverage decides.** The pre-trained vectors are as good as the share of *your* vocabulary they contain: news vectors miss the SMS tokens that matter.
- **In-domain wins** when you have plenty of unlabelled text from the *same* stream (a mail gateway has millions of messages).
- **Pre-trained wins** when your own text is tiny or the domain is close to the training text (product reviews, news).
- The third way is what the next sections show: **pre-train on so much text that coverage stops being a problem**, then keep the model frozen (sentence embeddings).

Words are made of pieces



- A word vector is the **sum of the vectors of its character n-grams** (Bojanowski et al. 2017): freee shares most n-grams with free.
- Skip-gram over words *and* n-grams gives every word a vector, **also unseen ones**. The supervised variant is average-then-linear (Joulin et al. 2017).

Out-of-vocabulary words in practice

Nearest neighbours of disguised words
(notebooks/01-spam/05):

word	trained on SMS	pre-trained, compressed
freee congratulationz	free , freemsg, freefone congratulations , congrats	guilty, dioxide (wrong) congratulations , Congrats
txtt	txtno, txt , txts	zero vector
fr33	fr, frm, fri (misses free)	zero vector

- Sub-words help, they do not guarantee: fr33 is dominated by its <fr n-gram. The 22 MB compressed model **pruned** the n-grams it would need.

Supervised fastText

- Same architecture, trained end to end on the labels (word bigrams + character n-grams, average, linear layer, softmax): seconds on a CPU.

Model	Precision	Recall	F1
fastText, mean of in-domain vectors + LR	0.975	0.907	0.940
fastText supervised	0.983	0.915	0.948
TF-IDF words + LR	0.983	0.907	0.944
TF-IDF characters + LR	0.983	0.899	0.939

- On clean text: **on par** with a tuned linear model. The gain is speed, dense vectors and tolerance to spelling variation, not a higher score.

Static versus contextual

- A static vector has **one meaning per word**: bank in river bank and in bank transfer share a vector; so do call in call now and call me later.
- A **Transformer encoder** reads the whole message and outputs a vector for each token **that depends on the rest of the message**.
- A **sentence embedding** pools those vectors (mean, or the last token) into **one vector per message**.
- **Small language model (SLM)**: a Transformer up to about 1 billion parameters, small enough for a laptop CPU or an integrated GPU.

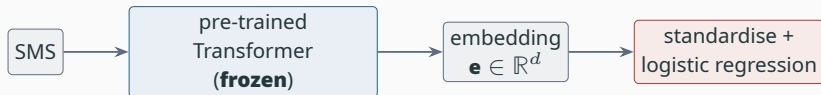
How an embedding model is trained

- **Contrastive learning** on billions of text pairs: pull a pair together, push the other texts of the batch apart (*InfoNCE* (Oord et al. 2018; Reimers and Gurevych 2019)).
- Similar meanings end up **close in cosine similarity**; normalised vectors: $\cos \theta = \mathbf{e}_1 \cdot \mathbf{e}_2$.

Pair of messages (EmbeddingGemma 2)	cosine
a prize message and a rewrite with other words	0.90
the prize message and a chat message	0.63
two random messages (mean of 200 pairs)	0.69

- TF-IDF: about 0 for the first pair. Compare cosines only **within one model** (MiniLM: 0.13 for random pairs).

Frozen encoder, linear probe



- Only $d + 1$ weights are trained: a *linear probe*.
Standardise each dimension first: LLM embeddings are **anisotropic** (all cosines high).
- No fine-tuning, no GPU for training. A small Keras MLP head (JAX) on the embeddings gives the same F1 (0.960): the vectors are already almost linearly separable.

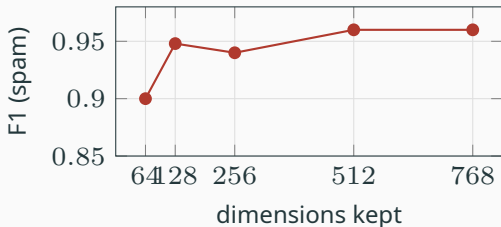
The 2026 zoo of small embedders

Model	Params	Dim.	4-bit file
all-MiniLM-L6-v2 (Reimers and Gurevych 2019)	22 M	384	21 MB
EmbeddingGemma 2 (Vera et al. 2025)	270 M	768	176 MB
Nomic Embed v2 MoE (Nussbaum and Duderstadt 2025)	475 M (305 M active)	768	344 MB

- Today: **EmbeddingGemma 2**, open weights, Matryoshka, 8k tokens, Apache-2.0. It matches the larger Nomic v2 (F1 0.96) in half the file and leads with few labels. English only.

Matryoshka: one vector, several sizes

The first d values of the vector form a valid smaller embedding (Kusupati et al. 2022). Keep them, normalise again. F1 of a linear probe (notebooks/01-spam/07):



- 512 dimensions lose nothing (2 KB per message); 256 and 128 stay within two points (one test message is 0.4 points: the dip is noise); 64 costs six.

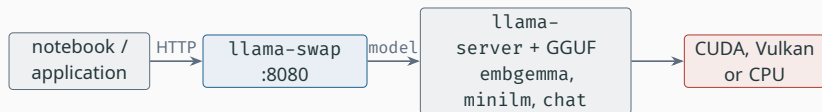
Does the task prefix matter?

EmbeddingGemma asks for title: none | text: in front of each document. F1 with a 95% bootstrap interval (1,000 resamples of the test set):

prefix	F1	95% interval
title: none text:	0.960	[0.932, 0.982]
none	0.960	[0.931, 0.983]
task: classification query:	0.948	[0.915, 0.975]

- The intervals overlap: **for a classification probe the prefix is irrelevant** here. It matters for retrieval (queries versus documents).
- Do not read a difference of 0.005 as an effect without an interval: 1,024 messages, about 130 spam.

Why a server?



- **Separation:** text in, vectors out; **no PyTorch, no TensorFlow**, no model in the notebook process; notebooks share one model.
- **Same API as commercial services** (`/v1/embeddings`, `/v1/chat/completions`).
- **llama.cpp** (Gerganov and contributors 2023): C++ engine for quantised models, on CPUs and on NVIDIA (CUDA) and most other GPUs (Vulkan).
- **llama-swap:** one endpoint; the `model` field picks the `llama-server`, loaded on demand.

GGUF and quantisation

- **GGUF**: one file with weights, vocabulary and metadata.
- **Quantisation**: weights in 8, 6, 5 or 4 bits (blocks share one scale) instead of 16: less disk and RAM, a little more error. UD- (*Unsloth Dynamic*): sensitive layers keep more bits.

File (EmbeddingGemma 2)	Size
BF16	558 MB
Q8_0	310 MB
UD-Q4_K_XL	176 MB

- Q4 vs Q8_0: cosine 0.993, F1 0.960 vs 0.964. **Every model of the course is 4 bit** (laptop RAM).

One image per GPU family

llama-swap publishes **unified images** (llama-swap + llama.cpp in one container). make llama-up picks the one that matches the GPU (detect_gpu.sh: nvidia-smi, then /dev/dri):

Profile	GPUs	Needs
cuda13	NVIDIA Ampere to Blackwell (RTX 30/40/50, A100, H100)	driver $\geq$ 580
cuda	NVIDIA Pascal to Ada (GTX 10xx, RTX 20xx)	CUDA 12
vulkan	AMD, Intel, any Vulkan GPU	/dev/dri
cpu	none (macOS, Windows)	Vulkan image, 0 GPU layers

- **Vulkan:** a cross-vendor graphics and compute API (AMD, Intel and NVIDIA drivers): no ROCm.

How fast?

Backend (messages/s, batches of 16)	speed
Vulkan, AMD Radeon 860M (integrated GPU)	55
CPU, same machine	27
MiniLM 22 M, CPU	195

- An integrated GPU only doubles the speed of the same machine's CPU (a discrete GPU is faster still; not measured here). Either way the SLM is a *second-stage* model.

The Docker Compose stack

```
services:
  llama-swap-vulkan:      # also: cuda13, cuda, cpu
    image: ghcr.io/mostlygeek/llama-swap:
      unified-vulkan
    devices: ["/dev/dri:/dev/dri"]
    volumes: ["/models:/models:ro", "/llama-swap.yaml: ... "]
    ports: ["127.0.0.1:8080:8080"]
```

- **Docker:** a program in a *container* (own file system);
Compose: containers, devices, files in one YAML file.
- **Floating tag** = latest llama.cpp (new model families need it); SWAP_TAG=<date> pins a build. Port bound to localhost.
- LLAMA_SWAP_LISTEN=0.0.0.0 inside, or the published port refuses connections.

Which model is loaded: llama-swap

```
models:
  embgemma:
    cmd: llama-server --port ${PORT} --embeddings --pooling mean
        -m /models/embeddinggemma-2-UD-Q4_K_XL.gguf -ngl ${env.NGL}
  minilm:
    cmd: llama-server --port ${PORT} --embeddings -ngl 0 ...
  chat:
    # Gemma 4 E2B, Q4
    cmd: llama-server --port ${PORT} -c 8192 ...
  ttl: 300
```

- `${PORT}`: chosen by llama-swap. **Policy** (matrix): the three models may share the memory (3 GB); by default only one.
- The **first request** after a swap waits while the GGUF loads. `-ngl`: layers on the GPU (99 = all, 0 with cpu); MiniLM stays on the CPU.

Getting the model

```
python download_model.py --repo unsloth/embeddinggemma-2-GGUF -  
-quant UD-Q4_K_XL  
python download_model.py --repo second-state/All-MiniLM-L6-v2-  
Embedding-GGUF --quant Q4_K_M
```

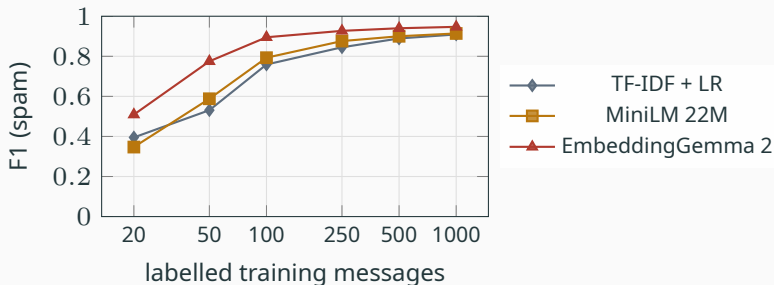
- EmbeddingGemma (176 MB) and MiniLM (21 MB); make `llama-models` fetches both (and the small chat model, 2.6 GB, for class 12). From Hugging Face; **resumable** (HTTP Range), **verified** (SHA-256 of the file against the Hub metadata), skips a file that is already good.
- A token for gated repositories comes from the environment (`HF_TOKEN`); it is never stored or printed.
- Pin what you run: image **tag**, model **file**, **checksum**. A latest tag is a supply-chain risk in a security course.

Talking to the server

```
POST http://127.0.0.1:8080/v1/embeddings
{"model": "embgemma", "input": ["title: none | text: <message>", ... ]}
```

- Response: `data[i].embedding`, 768 floats, **already L2-normalised**.
- Batch 16 to 64 messages per request; cache the vectors on disk (the model file name is part of the key).
- About 18 ms per message, batched or not: with `-np 1` the server handles them one after another. `"model": "minilm"` on the same URL reaches MiniLM.

Label efficiency of sentence embeddings



- Notebooks 06, 07 (10 to 20 repetitions). At 20 labels (3 spam) nothing is reliable; from **50** the embeddings lead (+24 points at 50, +5 at 500). Untuned TF-IDF scored 0.05 at 20 labels, tuned 0.40.

A new campaign: leave one family out

Spam falls in **families**. Train **without** one family, then test on it (recall; false positives below 1% for all models; families of 72 to 172 messages: suggestive, not proven):

held-out family	TF-IDF + LR	MiniLM + LR	EmbeddingGemma + LR
competition	0.695	0.562	0.848
prize claim	0.988	0.977	0.994
mobile upgrade	0.949	0.906	0.974
account statement	0.949	0.797	1.000
ringtones	0.889	0.889	0.847
chat lines	0.657	0.586	0.838
macro average	0.855	0.786	0.917

The price of meaning

Representation + classifier	F1	ms/msg	Size
TF-IDF + LR	0.944	0.04	about 1 MB
MiniLM (llama.cpp, CPU)	0.957	5.0	22 M parameters, 21 MB file
EmbeddingGemma 2 (llama.cpp, GPU)	0.960	18	270 M, 176 MB file

- With abundant labels the gap is **noise**; an SLM costs **125 to 450 times more** per message.
- **Design rule:** TF-IDF + linear first; an SLM as second stage, when labels are scarce or a new campaign appears.

The ladder in one table

Model	Precision	Recall	F1
Naive Bayes, counts	0.961	0.953	0.957
Logistic regression, TF-IDF	0.983	0.907	0.944
fastText supervised	0.983	0.915	0.948
Word2Vec skip-gram (in-domain) + LR	0.975	0.899	0.935
MiniLM + LR	0.976	0.938	0.957
EmbeddingGemma 2 + LR	0.984	0.938	0.960

- **All within 0.93 to 0.96.** The clean test score does not separate them. **Cost, label efficiency, new campaigns and robustness** do: the last one is next class.

Labs and practice guide

Notebooks in notebooks/01-spam/:

Notebook	Content
05	Word2Vec and fastText: in-domain vs pre-trained, OOV, few labels
06	SLM embeddings (MiniLM, EmbeddingGemma) through llama.cpp: probe, few labels, Keras head, cost
07	EmbeddingGemma 2 through llama.cpp: prefix, Matryoshka, new campaign, throughput
llama/	Docker Compose stack, llama-swap config, GGUF download script, smoke test

- **Practice guide** practice/spam/guide_02_02: word vectors, MiniLM, then your own llama.cpp stack (llama-swap). Fill-in notebook; solution published afterwards.

Take-aways

- **Vectors** add similarity to a bag of words: **in-domain** with plenty of unlabelled text from the same stream; **pre-trained** only if it covers your vocabulary.
- **Sentence embeddings** lead with few labels or a **new campaign**, at 125 to 450 times the cost.
- Run models as a **service** (llama.cpp behind llama-swap, GGUF, 4 bit, the image that matches your GPU); **pin** image, file and checksum.
- Report **intervals** and compare with a **fair baseline**.
- **Next class:** we attack every model on the ladder.

Word2Vec objective and sampling

For a corpus of pairs (w, c) with window L and noise distribution $P_n(w) \propto f(w)^{3/4}$ (Mikolov, Sutskever, et al. 2013):

$$J = - \sum_{(w,c)} \left[\log \sigma(\mathbf{v}_w \cdot \mathbf{u}_c) + \sum_{i=1}^k \mathbb{E}_{n_i \sim P_n} \log \sigma(-\mathbf{v}_w \cdot \mathbf{u}_{n_i}) \right]$$

- Frequent words are **sub-sampled**: a word w is dropped with probability $1 - \sqrt{t/f(w)}$, with $t \approx 10^{-5}$: fewer updates on the, to, you.
- Two matrices are learnt, V (word) and U (context); the word vectors used later are the rows of V .

Word2Vec factorises a PMI matrix

At the optimum of the negative-sampling objective (Levy and Goldberg 2014):

$$\mathbf{v}_w \mathbf{u}_c = \text{PMI}(w, c) - \log k, \quad \text{PMI}(w, c) = \log \frac{P(w, c)}{P(w)P(c)}$$

- Word2Vec is an implicit **low-rank factorisation** of the shifted word-context association matrix: the neighbourhood structure is a property of the co-occurrence counts, not of the neural network.
- With 65,000 tokens most pairs are seen once or never: the PMI estimates are noisy for rare words, hence the noisy neighbours.

Contrastive loss for sentence embeddings

For a batch of N pairs $(\mathbf{e}_i, \mathbf{e}_i^+)$ with cosine similarity s and temperature τ :

$$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(s(\mathbf{e}_i, \mathbf{e}_i^+)/\tau)}{\sum_j \exp(s(\mathbf{e}_i, \mathbf{e}_j^+)/\tau)}$$

- The other $N - 1$ texts of the batch are the **negatives**: larger batches give harder negatives (Oord et al. 2018).
- **Matryoshka loss** (Kusupati et al. 2022): the same loss summed over nested prefixes $\mathbf{e}_{1:d}$, $d \in \{64, 128, \dots, 768\}$, so that each prefix is a good embedding.

Mixture-of-experts routing (Nomic v2, not used today)

For a token state $\mathbf{x}$ and experts $E_1, \dots, E_8$ (Shazeer et al. 2017):

$$\mathbf{y} = \sum_{i \in T} g_i E_i(\mathbf{x}), \quad T = \text{top2}(\mathbf{W}_g \mathbf{x})$$

$$g = \text{softmax}((\mathbf{W}_g \mathbf{x})_T)$$

- Two feed-forward blocks run: compute like a smaller model, **memory** like the full one (Nomic v2: 475 M stored, 305 M used). A **load-balancing** term spreads the tokens over the experts.

Quantisation in one formula

A block of weights $\mathbf{w}$ is stored as integers q_i and one scale d (Q8_0: 8 bits):

$$d = \frac{\max_i |w_i|}{127}, \quad q_i = \text{round}(w_i/d), \quad \hat{w}_i = d q_i$$

- The error of every weight is at most $d/2$; with 8 bits it is below 0.4% of the largest weight of the block.
- Fewer bits (4-bit Q4 families): smaller file, larger error; for EmbeddingGemma 2 the mean cosine between the 4-bit and the 8-bit vector was 0.993 and the F1 0.960 against 0.964.

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